

# Measurement of Thermal Energy

## Form Three Revision Notes

---

### 1 Temperature, Thermal Energy 3 Heat Capacity and the Kinetic Picture

Every material is made of tiny particles (atoms or molecules) that are always moving. In a solid they vibrate about fixed positions; in a liquid they slide past one another; in a gas they fly about freely. Because they move, the particles carry *kinetic energy*, and because they attract one another they also store *potential energy*. The total of all this microscopic energy is the internal (thermal) energy of the body.

#### Definition

**Temperature** is a measure of the average kinetic energy of the particles of a body. It tells us how hot or cold a body is, but not how much heat energy the body contains. Its SI unit is the kelvin (K); everyday work often uses degrees Celsius ( $^{\circ}\text{C}$ ).

#### Definition

**Thermal (internal) energy** is the total kinetic and potential energy stored in all the particles of a body. It is a form of energy and is measured in joules (J).

#### NB

Temperature and heat are not the same thing. A cup of boiling water is hotter than a warm bath, yet the bath contains far more heat energy because it has a much greater mass. Convert temperatures with  $T/\text{K} = \theta/^{\circ}\text{C} + 273$ . A *change* of temperature has the same size in K and  $^{\circ}\text{C}$ , so  $\Delta\theta$  may be quoted in either unit.

### 2 Heat and What Determines It

When two bodies at different temperatures are placed together, energy flows from the hotter to the cooler one until they reach the same temperature. That energy in transit is called heat.

#### Definition

**Heat** is the energy transferred from one body to another because of a difference in temperature. Like all energy it is measured in joules (J).

The quantity of heat needed to warm a body depends on three things:

- the **mass** of the body (more mass needs more heat),
- the **temperature change** required (a bigger rise needs more heat),
- the **nature of the material** (different substances soak up heat differently).

### Definition

**Heat capacity**  $C$  of a body is the amount of heat energy needed to raise the temperature of the *whole body* by one kelvin (or one degree Celsius):

$$C = \frac{Q}{\Delta\theta}, \quad \text{unit: J K}^{-1}.$$

### NB

Heat capacity refers to a particular object, so a large kettle of water has a bigger heat capacity than a small cup of the same water. It does not tell us anything about the material on its own.

### Worked Example : heat capacity of a block

A heater delivers 9600 J of energy to an iron block and its temperature rises from 20 °C to 60 °C. Find the heat capacity of the block.

$$\Delta\theta = 60 - 20 = 40 \text{ K}$$
$$C = \frac{Q}{\Delta\theta} = \frac{9600 \text{ J}}{40 \text{ K}} = 240 \text{ J K}^{-1}.$$

### Worked Example : warming water

How much heat is needed to raise the temperature of 2.5 kg of water from 20 °C to 80 °C? Take  $c_{\text{water}} = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$ .

$$\Delta\theta = 80 - 20 = 60 \text{ K}$$
$$Q = m c \Delta\theta$$
$$= 2.5 \times 4200 \times 60$$
$$= 630\,000 \text{ J} = 630 \text{ kJ}.$$

### Worked Example : heating a metal block

An aluminium block of mass 0.8 kg is heated from 25 °C to 175 °C. Find the heat supplied. Take  $c_{\text{Al}} = 900 \text{ J kg}^{-1} \text{ K}^{-1}$ .

$$\Delta\theta = 175 - 25 = 150 \text{ K}$$
$$Q = m c \Delta\theta = 0.8 \times 900 \times 150$$
$$= 108\,000 \text{ J} = 108 \text{ kJ}.$$

### Worked Example : finding specific heat capacity

A 0.5 kg copper cylinder is given 7800 J of heat and its temperature rises by 40 K. Find its specific heat capacity.

$$c = \frac{Q}{m \Delta\theta} = \frac{7800}{0.5 \times 40}$$
$$= \frac{7800}{20} = 390 \text{ J kg}^{-1} \text{ K}^{-1}.$$

This matches the value for copper in the table above.

## 4 Specific Heat Capacity

### Definition

**Specific heat capacity**  $c$  of a substance is the amount of heat energy needed to raise the temperature of 1 kg of the substance by one kelvin:

$$c = \frac{Q}{m \Delta\theta}, \quad \text{unit: J kg}^{-1} \text{ K}^{-1}.$$

Rearranging gives the heat equation used constantly in this topic:

$$Q = m c \Delta\theta$$

### NB

Water has an unusually large specific heat capacity ( $4200 \text{ J kg}^{-1} \text{ K}^{-1}$ ). This is why water heats up slowly, cools down slowly and is used as a coolant in car engines and as a store of heat in hot-water systems.

#### Typical specific heat capacities

| Substance | $c \text{ (J kg}^{-1} \text{ K}^{-1})$ |
|-----------|----------------------------------------|
| Water     | 4200                                   |
| Ice       | 2100                                   |
| Steam     | 2010                                   |
| Aluminium | 900                                    |
| Iron      | 480                                    |
| Copper    | 390                                    |
| Brass     | 320                                    |
| Lead      | 130                                    |

## 5 Calorimetry and the Method of Mixtures

A **calorimeter** is a well-lagged metal container used to measure quantities of heat. When a hot body is placed with a cold one inside it, they exchange heat until they settle at a common (*equilibrium*) temperature.

### Definition

**The principle of mixtures** (conservation of energy): if no heat escapes to the surroundings, the heat lost by the hot body equals the heat gained by the cold body:

$$Q_{\text{lost}} = Q_{\text{gained}}$$

### NB

The container (calorimeter) and stirrer also absorb heat. In careful work their heat capacity is included. In these notes we assume a perfectly lagged calorimeter of negligible heat capacity unless stated otherwise.

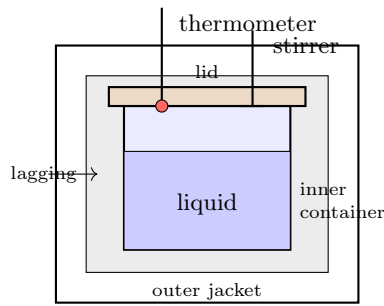


Fig. 1 A lagged calorimeter used in mixtures experiments.

#### Worked Example : equilibrium temperature

0.3 kg of water at 80 °C is mixed with 0.5 kg of water at 20 °C. Find the final temperature  $\theta$ .

Heat lost by hot water = heat gained by cold water:

$$\begin{aligned} m_1 c (80 - \theta) &= m_2 c (\theta - 20) \\ 0.3(80 - \theta) &= 0.5(\theta - 20) \\ 24 - 0.3\theta &= 0.5\theta - 10 \\ 34 &= 0.8\theta \\ \theta &= 42.5 \text{ } ^\circ\text{C}. \end{aligned}$$

#### Worked Example : specific heat capacity of a metal

A 0.4 kg metal block heated to 100 °C is dropped into 0.2 kg of water at 18 °C in a calorimeter of negligible heat capacity. The mixture settles at 30 °C. Find the specific heat capacity  $c_m$  of the metal.

$$\begin{aligned} Q_{\text{lost by metal}} &= Q_{\text{gained by water}} \\ m_m c_m (100 - 30) &= m_w c_w (30 - 18) \\ 0.4 c_m \times 70 &= 0.2 \times 4200 \times 12 \\ 28 c_m &= 10080 \\ c_m &= 360 \text{ J kg}^{-1} \text{ K}^{-1}. \end{aligned}$$

#### Worked Example : finding an unknown mass

An unknown mass  $m$  of water at 15 °C is added to 0.6 kg of water at 75 °C. The mixture reaches 55 °C. Find  $m$  (ignore the container).

$$\begin{aligned} m c (55 - 15) &= 0.6 c (75 - 55) \\ 40 m &= 0.6 \times 20 \\ 40 m &= 12 \\ m &= 0.3 \text{ kg}. \end{aligned}$$

#### Worked Example : specific heat capacity of a liquid

A heater rated 50 W warms 0.25 kg of an oil by 16 K in 2 min. Assuming no losses, find the specific heat capacity of the oil.

$$\begin{aligned} Q &= P t = 50 \times (2 \times 60) = 6000 \text{ J} \\ c &= \frac{Q}{m \Delta \theta} = \frac{6000}{0.25 \times 16} \\ &= \frac{6000}{4} = 1500 \text{ J kg}^{-1} \text{ K}^{-1}. \end{aligned}$$

## 6 Change of State and the Kinetic Theory

Matter exists in three common states: solid, liquid and gas. Adding or removing heat can change one state into another. The kinetic theory explains this by how strongly the particles are held together and how fast they move.

#### Definition

**Melting** is the change from solid to liquid at a fixed temperature called the **melting point**. **Freezing** is the reverse change from liquid to solid at the same temperature (the **freezing point**). For a pure substance the two are equal.

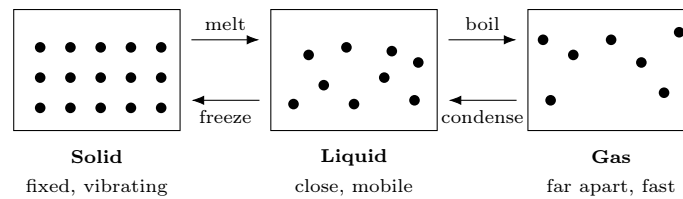
#### NB

While a pure solid melts, the temperature stays constant even though heat is still being supplied. The energy goes into breaking the bonds between particles, not into raising their speed.

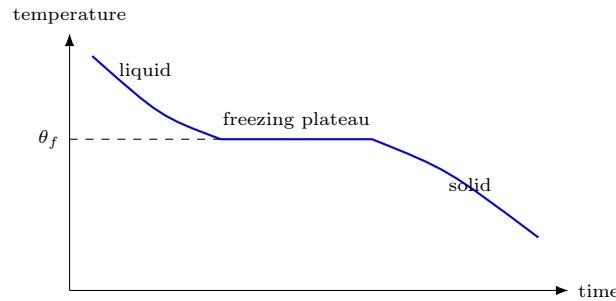
#### Worked Example : cooling through the freezing point

A pure substance of mass 0.2 kg is liquid at 90 °C and freezes at 60 °C. Its specific heat capacity as a liquid is 2500 J kg<sup>-1</sup> K<sup>-1</sup> and its specific latent heat of fusion is 80 000 J kg<sup>-1</sup>. Find the total heat removed to cool the liquid to 60 °C and then freeze it completely.

$$\begin{aligned} Q_1 &= m c \Delta \theta = 0.2 \times 2500 \times (90 - 60) \\ &= 15\,000 \text{ J} \\ Q_2 &= m L_f = 0.2 \times 80\,000 = 16\,000 \text{ J} \\ Q_{\text{total}} &= Q_1 + Q_2 = 31\,000 \text{ J} = 31 \text{ kJ}. \end{aligned}$$



**Fig. 2** The three states of water and the changes between them.



**Fig. 3** Cooling curve: a liquid cooling to a solid shows a flat plateau at the freezing point  $\theta_f$ .

## 7 Effect of Impurities and Pressure; Regelation

- **Impurities** lower the melting/freezing point. Salt spread on icy roads makes the ice melt below  $0^\circ\text{C}$ .
- **Increasing pressure lowers** the melting point of ice (ice is unusual because it expands on freezing). For most substances higher pressure raises the melting point.

### Definition

**Regelation** is the melting of ice under pressure and its re-freezing once the pressure is removed. A thin wire loaded with weights slowly passes through a block of ice while the block stays in one piece behind it.

## 8 Boiling and Evaporation

Both boiling and evaporation turn a liquid into a gas, but they are not the same process.

### Definition

**Boiling** is the rapid change of a liquid to vapour throughout the whole liquid at a fixed temperature, the **boiling point**. **Evaporation** is the slow escape of the fastest molecules from the surface of a liquid, and it happens at all temperatures.

| Boiling                       | Evaporation                  |
|-------------------------------|------------------------------|
| Occurs at a fixed temperature | Occurs at any temperature    |
| Throughout the liquid         | Only at the surface          |
| Bubbles form in the bulk      | No bubbles                   |
| Needs continuous heating      | Can occur without added heat |

The boiling point is affected by:

- **Impurities:** dissolved substances raise the boiling point (salty water boils above  $100^\circ\text{C}$ ).
- **Pressure:** higher pressure raises the boiling point (a pressure cooker cooks faster). Lower pressure lowers it.

- **Altitude:** on high mountains the air pressure is low, so water boils below  $100^\circ\text{C}$ .

## 9 Latent Heat

During a change of state heat is absorbed or released with *no change in temperature*. This “hidden” heat is called latent heat.

### Definition

**Specific latent heat of fusion**  $L_f$  is the heat needed to change 1 kg of a solid into liquid at its melting point without a change in temperature. **Specific latent heat of vapourisation**  $L_v$  is the heat needed to change 1 kg of a liquid into vapour at its boiling point. In both cases:

$$Q = mL, \quad \text{unit of } L : \text{J kg}^{-1}.$$

### NB

For water:  $L_f = 3.34 \times 10^5 \text{ J kg}^{-1}$  and  $L_v = 2.26 \times 10^6 \text{ J kg}^{-1}$ . Vapourisation absorbs far more energy than fusion, which is why steam at  $100^\circ\text{C}$  causes worse scalds than the same mass of boiling water.

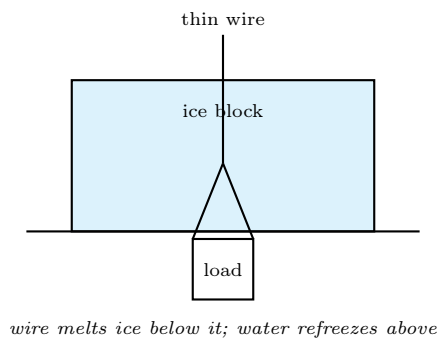
### Worked Example : melting ice

Find the heat needed to melt 0.4 kg of ice already at  $0^\circ\text{C}$ . Take  $L_f = 3.34 \times 10^5 \text{ J kg}^{-1}$ .

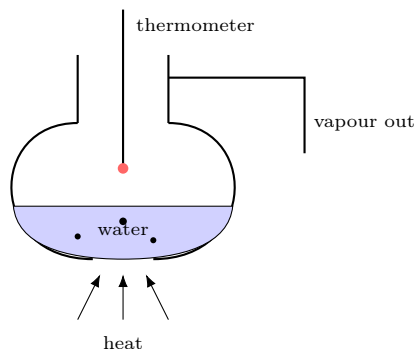
$$\begin{aligned}
 Q &= mL_f = 0.4 \times 3.34 \times 10^5 \\
 &= 133\,600 \text{ J} \approx 134 \text{ kJ}.
 \end{aligned}$$

### Worked Example : ice to steam in stages

Find the total heat to change 0.1 kg of ice at  $-10^\circ\text{C}$  into steam at  $120^\circ\text{C}$ . Use  $c_{\text{ice}} = 2100$ ,  $c_{\text{water}} = 4200$ ,  $c_{\text{steam}} = 2010 \text{ (J kg}^{-1} \text{ K}^{-1})$ ,  $L_f = 3.34 \times 10^5 \text{ J kg}^{-1}$ ,  $L_v = 2.26 \times 10^6 \text{ J kg}^{-1}$ .



**Fig. 4** Regelation: a loaded wire cuts through ice which re-freezes behind it.



**Fig. 5** Boiling apparatus for observing the boiling point.

$$\begin{aligned}
 Q_1 &= m_{\text{ice}} \Delta\theta = 0.1 \times 2100 \times 10 = 2100 \text{ J} \\
 Q_2 &= mL_f = 0.1 \times 3.34 \times 10^5 = 33\,400 \text{ J} \\
 Q_3 &= mc_w \Delta\theta = 0.1 \times 4200 \times 100 = 42\,000 \text{ J} \\
 Q_4 &= mL_v = 0.1 \times 2.26 \times 10^6 = 226\,000 \text{ J} \\
 Q_5 &= mc_{\text{st}} \Delta\theta = 0.1 \times 2010 \times 20 = 4020 \text{ J} \\
 Q_{\text{tot}} &= Q_1 + Q_2 + Q_3 + Q_4 + Q_5 \\
 &= 307\,520 \text{ J} \approx 308 \text{ kJ}.
 \end{aligned}$$

$$\begin{aligned}
 Q &= mL_f = 0.5 \times 25000 = 12\,500 \text{ J} \\
 t &= \frac{Q}{P} = \frac{12500}{300} \approx 41.7 \text{ s}.
 \end{aligned}$$

### Worked Example : steam condensing to warm water

Steam at  $100^\circ\text{C}$  is passed into  $0.5 \text{ kg}$  of water at  $20^\circ\text{C}$  until the water reaches  $40^\circ\text{C}$ . The condensed steam then cools from  $100^\circ\text{C}$  to  $40^\circ\text{C}$ . Find the mass  $m_s$  of steam condensed. Use  $L_v = 2.26 \times 10^6 \text{ J kg}^{-1}$ ,  $c_w = 4200$ .

Heat released by steam = heat gained by cold water:

$$\begin{aligned}
 m_s L_v + m_s c_w (100 - 40) &= m_w c_w (40 - 20) \\
 m_s (2.26 \times 10^6 + 4200 \times 60) &= 0.5 \times 4200 \times 20 \\
 m_s (2.26 \times 10^6 + 2.52 \times 10^5) &= 42000 \\
 m_s \times 2.512 \times 10^6 &= 42000 \\
 m_s &\approx 0.0167 \text{ kg} = 16.7 \text{ g}.
 \end{aligned}$$

### Worked Example : energy and time to melt lead

A heater rated  $300 \text{ W}$  melts  $0.5 \text{ kg}$  of lead already at its melting point. Take  $L_f(\text{lead}) = 25\,000 \text{ J kg}^{-1}$ . Find the energy required and the time taken, assuming no heat is lost.

## 10 Cooling by Evaporation and the Refrigerator

When a liquid evaporates, the fastest molecules escape, so the average kinetic energy of those left behind falls. This is why evaporation cools: sweat drying on the skin, or a wet cloth round a water pot, both feel cold. Volatile liquids (methylated spirit, ether) evaporate fast and produce strong cooling.

### Definition

A **refrigerator** uses a volatile working fluid (refrigerant) that evaporates inside the freezer coils, taking latent heat from the food, and then condenses in the coils behind the fridge, giving that heat out to the room.

### NB

The pump keeps the refrigerant circulating. The refrigerant *evaporates* in the cold part (absorbing heat) and *condenses* in the warm part (releasing heat), so heat is pumped from inside the fridge to the room.

## 11 Graph-Reading Practice

Fig. 10 shows a solid being heated steadily until it becomes a gas. Regions AB and CD are sloping (temperature rising) while BC and DE are flat (change of state). Reading such

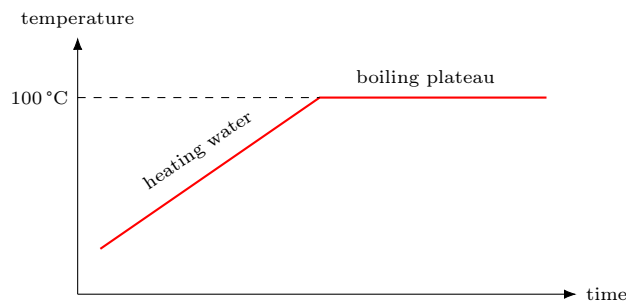


Fig. 6 Heating water: the temperature rises then stays flat while the water boils.

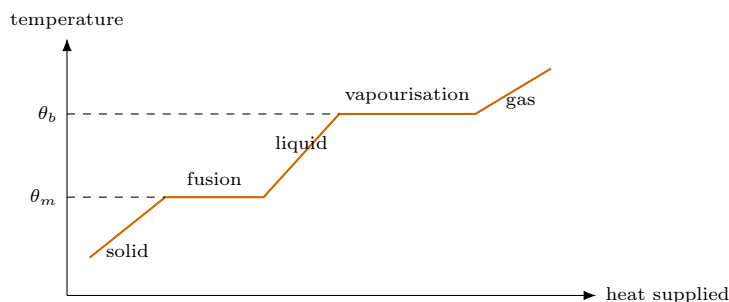


Fig. 7 Temperature versus heat supplied: two flat plateaus mark fusion and vapourisation.

a graph: sloping parts use  $Q = mc\Delta\theta$ ; flat plateaus use table on page 1.  
 $Q = mL$ .

### Chapter Summary

- **Temperature** measures the average kinetic energy of particles; **heat** is energy transferred because of a temperature difference. Both differ in meaning.
- Heat supplied depends on mass, temperature change and the nature of the material.
- **Heat capacity**  $C = Q/\Delta\theta$  ( $\text{J K}^{-1}$ ) applies to a whole object.
- **Specific heat capacity**  $c = Q/(m \Delta\theta)$ ; use  $Q = mc \Delta\theta$ . Water's large  $c$  ( $4200 \text{ J kg}^{-1} \text{ K}^{-1}$ ) makes it a good coolant and heat store.
- **Method of mixtures**: in a lagged calorimeter, heat lost = heat gained.
- Matter has three states; the **kinetic theory** explains state changes by particle spacing and speed.
- **Melting/boiling** happen at fixed temperatures; the temperature stays constant during a change of state.
- **Latent heat of fusion**  $L_f$  (solid→liquid) and **vapourisation**  $L_v$  (liquid→gas):  $Q = mL$ ; for water  $L_v \gg L_f$ .
- **Impurities** lower the melting point but raise the boiling point; **pressure** lowers ice's melting point (regelation) and raises the boiling point.
- **Evaporation cools**; a refrigerator uses evaporation and condensation of a refrigerant to pump heat out of the cabinet.

### Part A — Conceptual / short answer

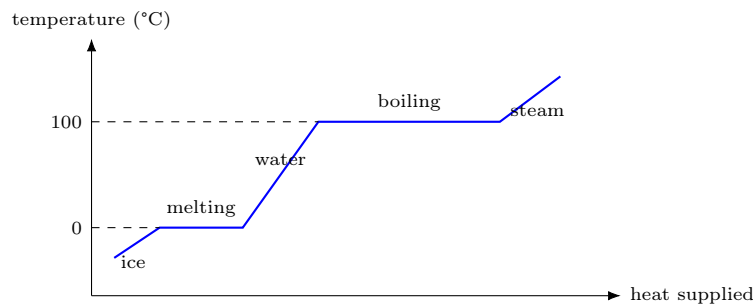
- Q1. Explain, using the kinetic theory, why standing in a breeze with wet skin makes you feel cold.
- Q2. Give two reasons why water is chosen as the coolant in a car engine.
- Q3. State three differences between boiling and evaporation.
- Q4. List three factors that change the boiling point of a liquid, and say in which direction each one moves it.
- Q5. Describe three ways of speeding up the evaporation of a wet surface.
- Q6. Explain why a drink is cooled more effectively by a lump of ice at  $0^\circ\text{C}$  than by the same mass of water at  $0^\circ\text{C}$ .
- Q7. A loaded wire slowly cuts through a block of ice, yet the block stays whole. Name this effect and explain it.
- Q8. Outline how a refrigerator uses a volatile fluid to move heat from the food inside it to the room.
- Q9. State the effect of dissolving salt in water on (i) its freezing point and (ii) its boiling point.

### Part B — Calculations using Q

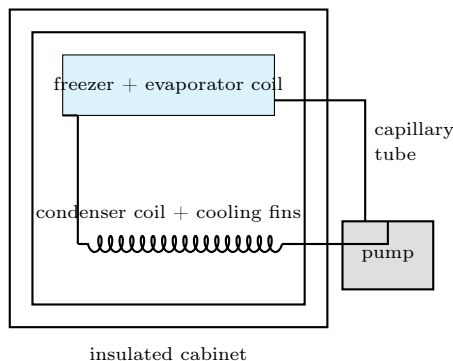
- Q10. How much heat is needed to raise the temperature of 3 kg of water from  $15^\circ\text{C}$  to  $65^\circ\text{C}$ ?
- Q11. An aluminium saucepan of mass 1.5 kg is heated from  $20^\circ\text{C}$  to  $120^\circ\text{C}$ . Find the heat absorbed. ( $c_{\text{Al}} = 900$ .)
- Q12. The same 12 000 J of heat is given separately to 0.5 kg of water and to 0.5 kg of iron ( $c_{\text{iron}} = 480$ ). Find the temperature rise of each and comment on the difference.

## 12 Practice Questions

Work through these on your own, then check your method against the **Solutions** section at the end of the notes. Take  $c_{\text{water}} = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$ ,  $c_{\text{ice}} = 2100 \text{ J kg}^{-1} \text{ K}^{-1}$ ,  $c_{\text{steam}} = 2010 \text{ J kg}^{-1} \text{ K}^{-1}$ ,  $L_f(\text{ice}) = 3.34 \times 10^5 \text{ J kg}^{-1}$ ,  $L_v(\text{water}) = 2.26 \times 10^6 \text{ J kg}^{-1}$ , and other  $c$  values from the



**Fig. 8** Staged heating of ice at  $-10^{\circ}\text{C}$  to steam at  $120^{\circ}\text{C}$ .



**Fig. 9** Domestic refrigerator: evaporator coil in the freezer, capillary tube, compressor pump and condenser fins.

### Part C — Method of mixtures

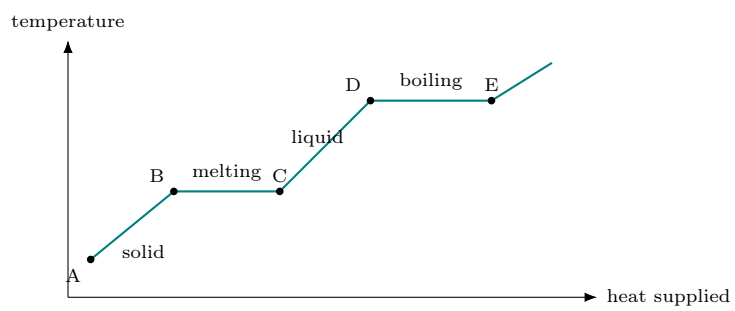
- Q13.** 0.4 kg of water at  $70^{\circ}\text{C}$  is mixed with 0.6 kg of water at  $20^{\circ}\text{C}$ . Find the final temperature (ignore the container).
- Q14.** A 0.5 kg copper block ( $c = 390$ ) heated to  $95^{\circ}\text{C}$  is dropped into 0.3 kg of water at  $22^{\circ}\text{C}$  in a calorimeter of negligible heat capacity. Find the final temperature.
- Q15.** An aluminium cup of mass 0.15 kg ( $c = 900$ ) holds 0.25 kg of water, all at  $18^{\circ}\text{C}$ . A 0.2 kg iron block ( $c = 480$ ) at  $150^{\circ}\text{C}$  is lowered in. Find the common temperature reached (assume no heat is lost).

### Part D — Latent heat

- Q16.** Find the heat needed to melt 0.25 kg of ice that is already at  $0^{\circ}\text{C}$ .
- Q17.** Find the heat needed to turn 0.2 kg of ice at  $0^{\circ}\text{C}$  into water at  $30^{\circ}\text{C}$ .
- Q18.** How much heat must be removed to freeze 0.5 kg of water already at  $0^{\circ}\text{C}$ ?
- Q19.** A 500 W immersion heater is switched on for 2 min in a mixture of ice and water at  $0^{\circ}\text{C}$ . What mass of ice melts (no losses)?
- Q20.** Find the heat needed to change 0.3 kg of water at  $50^{\circ}\text{C}$  into steam at  $100^{\circ}\text{C}$ .
- Q21.** A refrigerator freezes 0.6 kg of water already at  $0^{\circ}\text{C}$  into ice at  $0^{\circ}\text{C}$  in 4 min. Find the average rate at which it removes heat.

### Part E — Graph reading

- Q22.** 0.5 kg of a pure solid is heated steadily and gives the graph of Fig. 10. From the graph, the sloping part AB (solid warming) runs from  $0^{\circ}\text{C}$  to  $50^{\circ}\text{C}$  and absorbs 21 000 J; the flat part BC is at  $50^{\circ}\text{C}$  and the flat part DE is at  $120^{\circ}\text{C}$ ; the boiling plateau DE absorbs 565 000 J.
- Write down the melting point and the boiling point.
  - Find the specific heat capacity of the solid.
  - Find the specific latent heat of vapourisation.



**Fig. 10** Generic heating graph for a solid taken through melting and boiling.



## Solutions to Practice Questions

**Standard constants used throughout:**  $c_{\text{water}} = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$ ,  $c_{\text{ice}} = 2100 \text{ J kg}^{-1} \text{ K}^{-1}$ ,  $c_{\text{steam}} = 2010 \text{ J kg}^{-1} \text{ K}^{-1}$ ,  $c_{\text{Al}} = 900 \text{ J kg}^{-1} \text{ K}^{-1}$ ,  $c_{\text{iron}} = 480 \text{ J kg}^{-1} \text{ K}^{-1}$ ,  $c_{\text{copper}} = 390 \text{ J kg}^{-1} \text{ K}^{-1}$ ,  $L_f(\text{ice}) = 3.34 \times 10^5 \text{ J kg}^{-1}$ ,  $L_v(\text{water}) = 2.26 \times 10^6 \text{ J kg}^{-1}$ .

### Part A — Conceptual

**Q1.** Water on the skin evaporates, and it is the fastest (most energetic) molecules that escape. This lowers the average kinetic energy of the water and skin left behind, so their temperature falls. A breeze sweeps the vapour away, keeping the escape going, so the cooling is faster and you feel colder.

**Q2.** (i) Water has a very large specific heat capacity, so it can carry away a lot of heat for only a small rise in its own temperature. (ii) It is cheap, plentiful and easy to pump around the engine. (Its high latent heat and non-flammability are also acceptable reasons.)

**Q3.** Boiling occurs at one fixed temperature, throughout the whole liquid, with bubbles forming in the bulk and needs a steady supply of heat. Evaporation occurs at any temperature, only at the surface, forms no bubbles and can happen with no added heat.

**Q4.** (i) Dissolved impurities *raise* the boiling point. (ii) Increased external pressure *raises* the boiling point (lower pressure lowers it). (iii) Higher altitude means lower pressure, which *lowers* the boiling point.

**Q5.** Increase the surface area (spread the liquid out), raise the temperature, and blow a draught of dry air across it to carry the vapour away. (Reducing the surrounding pressure or humidity also works.)

**Q6.** Both are at  $0^\circ\text{C}$ , but the ice must first absorb its latent heat of fusion ( $3.34 \times 10^5 \text{ J kg}^{-1}$ ) from the drink in order to melt, before the melt-water warms up. The water at  $0^\circ\text{C}$  takes only the much smaller  $Q = mc\Delta\theta$  as it warms. So the ice removes far more heat from the drink and cools it more.

**Q7.** This is **regelation**. The pressure of the wire lowers the melting point of the ice directly beneath it, so it melts; the melt-water flows above the wire where the pressure is normal, and re-freezes, giving out latent heat. The wire moves down while the block stays in one piece.

**Q8.** A pump circulates a volatile refrigerant. In the evaporator coils inside the cabinet the refrigerant evaporates, absorbing its latent heat from the food and cooling it. The vapour is pumped to the condenser coils behind the fridge, where it is compressed and condenses, releasing that latent heat to the room. The cycle repeats, pumping heat from inside to outside.

**Q9.** Adding salt (i) *lowers* the freezing point of the water and (ii) *raises* its boiling point.

### Part B — $Q = mc\Delta\theta$

**Q10.**

$$\begin{aligned}\Delta\theta &= 65 - 15 = 50 \text{ K} \\ Q &= mc\Delta\theta = 3 \times 4200 \times 50 \\ &= 630\,000 \text{ J} = 630 \text{ kJ}.\end{aligned}$$

**Q11.**

$$\begin{aligned}\Delta\theta &= 120 - 20 = 100 \text{ K} \\ Q &= mc\Delta\theta = 1.5 \times 900 \times 100 \\ &= 135\,000 \text{ J} = 135 \text{ kJ}.\end{aligned}$$

**Q12.** Using  $\Delta\theta = Q/(mc)$  with  $Q = 12\,000 \text{ J}$  and  $m = 0.5 \text{ kg}$ :

$$\begin{aligned}\Delta\theta_{\text{water}} &= \frac{12000}{0.5 \times 4200} = \frac{12000}{2100} \approx 5.7 \text{ K} \\ \Delta\theta_{\text{iron}} &= \frac{12000}{0.5 \times 480} = \frac{12000}{240} = 50 \text{ K}.\end{aligned}$$

The iron warms about  $50/5.7 \approx 8.75$  times as much, because its specific heat capacity is much smaller: the same heat causes a far bigger temperature rise in iron than in water.

## Part C — Method of mixtures

**Q13.** Heat lost by hot water = heat gained by cold water (the  $c$  cancels):

$$\begin{aligned}0.4(70 - \theta) &= 0.6(\theta - 20) \\28 - 0.4\theta &= 0.6\theta - 12 \\40 &= 1.0\theta \\ \theta &= 40^\circ\text{C}.\end{aligned}$$

This lies between  $20$  and  $70^\circ\text{C}$ , as expected.

**Q14.** Heat lost by copper = heat gained by water:

$$\begin{aligned}0.5 \times 390 \times (95 - \theta) &= 0.3 \times 4200 \times (\theta - 22) \\195(95 - \theta) &= 1260(\theta - 22) \\18525 - 195\theta &= 1260\theta - 27720 \\46245 &= 1455\theta \\ \theta &\approx 31.8^\circ\text{C}.\end{aligned}$$

This lies between  $22$  and  $95^\circ\text{C}$ , as expected.

**Q15.** Heat lost by iron = heat gained by (cup + water):

$$\begin{aligned}0.2 \times 480 \times (150 - \theta) &= (0.15 \times 900 + 0.25 \times 4200)(\theta - 18) \\96(150 - \theta) &= (135 + 1050)(\theta - 18) \\96(150 - \theta) &= 1185(\theta - 18) \\14400 - 96\theta &= 1185\theta - 21330 \\35730 &= 1281\theta \\ \theta &\approx 27.9^\circ\text{C}.\end{aligned}$$

This lies between  $18$  and  $150^\circ\text{C}$ , as expected.

## Part D — Latent heat

**Q16.**

$$\begin{aligned}Q &= mL_f = 0.25 \times 3.34 \times 10^5 \\ &= 83\,500\text{ J} = 83.5\text{ kJ}.\end{aligned}$$

**Q17.** Melt the ice, then warm the melt-water:

$$\begin{aligned}Q_1 &= mL_f = 0.2 \times 3.34 \times 10^5 = 66\,800\text{ J} \\Q_2 &= mc_w \Delta\theta = 0.2 \times 4200 \times 30 = 25\,200\text{ J} \\Q &= Q_1 + Q_2 = 92\,000\text{ J} = 92\text{ kJ}.\end{aligned}$$

**Q18.** Freezing removes the latent heat of fusion:

$$\begin{aligned}Q &= mL_f = 0.5 \times 3.34 \times 10^5 \\ &= 167\,000\text{ J} = 167\text{ kJ removed}.\end{aligned}$$

**Q19.** Energy supplied melts the ice ( $Q = mL_f$ ):

$$\begin{aligned}Q &= Pt = 500 \times (2 \times 60) = 60\,000\text{ J} \\m &= \frac{Q}{L_f} = \frac{60000}{3.34 \times 10^5} \\ &\approx 0.180\text{ kg} = 180\text{ g}.\end{aligned}$$

**Q20.** Warm the water to  $100^\circ\text{C}$ , then boil it:

$$\begin{aligned}Q_1 &= mc_w \Delta\theta = 0.3 \times 4200 \times 50 = 63\,000\text{ J} \\Q_2 &= mL_v = 0.3 \times 2.26 \times 10^6 = 678\,000\text{ J} \\Q &= Q_1 + Q_2 = 741\,000\text{ J} = 741\text{ kJ}.\end{aligned}$$

**Q21.** Heat that must be removed to freeze the water:

$$\begin{aligned}Q &= mL_f = 0.6 \times 3.34 \times 10^5 = 200\,400 \text{ J} \\t &= 4 \times 60 = 240 \text{ s} \\ \text{rate} &= \frac{Q}{t} = \frac{200400}{240} \approx 835 \text{ W}.\end{aligned}$$

## Part E — Graph reading

**Q22.** (a) The flat parts mark the changes of state, so the **melting point** is  $50^\circ\text{C}$  (plateau BC) and the **boiling point** is  $120^\circ\text{C}$  (plateau DE).

(b) Segment AB warms the solid from  $0$  to  $50^\circ\text{C}$  using  $Q = mc\Delta\theta$ :

$$\begin{aligned}c &= \frac{Q}{m\Delta\theta} = \frac{21000}{0.5 \times 50} \\ &= \frac{21000}{25} = 840 \text{ J kg}^{-1} \text{ K}^{-1}.\end{aligned}$$

(c) Plateau DE is the boiling (vapourisation) stage, using  $Q = mL_v$ :

$$\begin{aligned}L_v &= \frac{Q}{m} = \frac{565000}{0.5} \\ &= 1.13 \times 10^6 \text{ J kg}^{-1}.\end{aligned}$$